

week 2

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Report: Word Embeddings and LSTM-based Sentiment Analysis

Objective

The objective of this experiment was to understand distributed word representations and their application in deep learning models. The task involved training word embeddings using Word2Vec or GloVe, building an LSTM-based sentiment classifier, and visualizing word embeddings using dimensionality reduction techniques.

Methodology

A small text corpus was first tokenized and used to train Word2Vec embeddings, which capture semantic relationships between words in a continuous vector space. Pretrained GloVe embeddings were also explored for comparison. For sentiment analysis, the IMDb movie review dataset was used. Text data was tokenized, converted into sequences, and padded to a fixed length. An LSTM-based neural network was then trained to classify movie reviews as positive or negative, leveraging sequential dependencies in text.

To analyze semantic similarity among words, trained word vectors were visualized using Principal Component Analysis (PCA) and t-SNE, which project high-dimensional embeddings into two dimensions.

Conclusion

The experiment demonstrated the effectiveness of word embeddings and LSTM models in capturing semantic and contextual information for sentiment classification, providing a foundation for advanced NLP models.